

CUTANEOUS SIGNS AND SYMPTOMS ASSOCIATED WITH HEMATOLOGIC DISEASES

Pallor

Pallor of the mucocutaneous tissue is most readily observed in the palms, nail beds, face, and conjunctivae—areas where the microvascular bed is visible through the skin. It results from decreased hemoglobin content.

Multiple factors can cause pallor, including: - Acute and chronic hemorrhage - Emotional stress - Vasopressor drugs - Cold exposure - Shock - Pathophysiologic conditions leading to anemia - Cutaneous edema

Addressing the underlying cause typically resolves the pallor.

Anemia

The palmar creases serve as a useful indicator of anemia severity. Normally, they appear pink unless the hemoglobin level falls below the normal range.

Anemia is defined based on the patient's age and sex. A morphologic classification using red blood cell (RBC) size or volume—assessed via blood smear—helps categorize anemias into: - **Microcytic** (small RBCs) - **Normocytic** (normal-sized RBCs) - **Macrocytic** (large RBCs)

Each type correlates with specific etiologies such as: - Blood loss - Genetic disorders - Decreased RBC production - Increased RBC destruction

Constitutional Signs and Symptoms

Common systemic manifestations of anemia include: - Malaise - Fatigue - Dizziness - Postural hypotension - Dyspnea on exertion - Tachycardia

These symptoms become more pronounced with increased oxygen demand. Red blood cells transport oxygen bound to hemoglobin from the lungs to tissue capillaries. Oxygen release is governed by the characteristics of the oxyhemoglobin dissociation curve.

Laboratory evaluation should be guided by clinical suspicion and typically includes: - Complete blood count with white blood cell differential - Iron studies (serum iron, total iron-binding capacity, ferritin)

HEMATOLOGIC DISEASES AT A GLANCE

Mucocutaneous changes—such as pallor, jaundice, flushing, erythema, and cyanosis—can signal underlying hematologic disorders.

Key clinical points: - Anemia is best classified by mean corpuscular volume (MCV). - Megaloblastic anemia due to vitamin B₁₂ deficiency is distinguished from folate deficiency by the presence of neurologic symptoms (absent in folate deficiency). - Flushing in carcinoid tumor typically lasts 10 minutes or less; in mastocytosis, it lasts 30 minutes or more. - Polycythemia vera presents with acrocyanosis, aquagenic pruritus, urticaria, Sweet syndrome, and purpura. - The morphologic approach to purpura involves the three P's:

1. Is the lesion purpuric?
2. Is it primary?
3. Is it palpable? - Leukemia cutis refers to localized or disseminated skin infiltration by leukemic cells, indicating systemic disease, dissemination, or relapse.